Problem #1

```
FUNCTION solution(A, K)
    SET n to the length of A

    IF n is 0 THEN
        RETURN A
    END IF

    Create a new list called result with n spots

    FOR each position i from 0 to n-1
        SET newPosition to the remainder of (i+K) divided by n
        PUT A[i] into result of newPosition
    ENDFOR

    RETURN result
END FUNCTION
```

Problem #2

```
FUNCTION solution(A)
    CREATE an empty set named unpaired

    FOR each number in A
        IF number is in unpaired THEN
            REMOVE number from unpaired
        ELSE
            ADD number to unpaired
        END IF
    END FOR

    RETURN the one value left in unpaired
END FUNCTION
```

Problem #3

```
FUNCTION solution(X, Y, D)
    SET distance to Y-X
    SET jumps to (distance + D-1) divided by D, drop remainder
    RETURN jumps
END FUNCTION
```

problem #4

```
FUNCTION solution(A)
    SET n to the length of A
    CREATE a set called seen containing every number in A

    FOR each number from 1 to n+1
        IF number is not in seen THEN
            RETURN number
        END IF
    END FOR
END FUNCTION
```

Problem #5

```
FUNCTION solution(A)
    SET total to the sum of all numbers in A
    SET left to 0
    SET minDiff to a big number

    FOR each split part P from 1 to -1
        ADD A(P-1) to left
        SET right to total - left
        SET diff to |(left - right)|

        IF diff is < minDiff THEN
            SET minDiff to diff
        END IF
    END FOR

    RETURN minDiff
END FUNCTION
```